

Ruben Francisco Carbajal

713-826-3476 • ruben.carbajal6502@gmail.com • linkedin/rubenfc304 • github/Ruben304 • Personal-Website

EXPERIENCE

JP Morgan Chase & Co | Software Engineer

Aug 2024- Present

- Built a self-service data exploration tool letting users slice and dice records across upstream and downstream sources, designing the React interface, Flask REST endpoints, and CockroachDB schemas that both joined previously siloed datasets and persisted user-defined queries and saved views
- Extended the platform with a visualization layer offering 6 configurable chart views over the same data, adding in-app data manipulation and a cross-team sharing feature for saved queries and views; adopted by 7 teams
- Helped decouple a critical workflow from a third-party application dependency by rebuilding it in house, contributing to backend architecture, React UI, and a file transfer channel for automated data exchange between systems
- Contributed to a full modernization of a legacy application, migrating from an outdated Oracle database and in-house custom framework to CockroachDB and industry-standard Flask libraries, refactoring legacy code and porting 14 years of production data with zero downtime

JP Morgan Chase & Co | Software Engineering Intern

Jun 2023 - Aug 2023

- Automated two internal reporting services used by ~110K employees firm-wide, cutting a manual ~15-minute workflow to under ~2 minutes and eliminating daily manual task
- Built the system end to end: a React/JavaScript interface, Spring Boot REST endpoints with server-side request validation and error handling, and Maven-built Java services backed by AWS
- Authored Groovy-based Jenkins pipelines automating build, test, and deployment, and presented the finished tool to [N] stakeholder teams before the end of the 10-week internship

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C, C++, Java, Groovy

AI: GitHub Copilot, Prompt Engineering, AI Agents, Skills Building, LLM API Integration

Frameworks & Libraries: React, Vue.js, Flask, SQLAlchemy, Pydantic, Pytest, Spring Boot, Cypress, Vite, AG Grid, Highcharts

Databases & Tools: PostgreSQL, CockroachDB, MongoDB, Oracle, Git, Jenkins, SonarQube, Poetry, Maven, Jira/Agile

PROJECTS

VIAD Optics

[GitHub](#)

- Led the computer-vision pipeline for assistive smart glasses (5-person capstone), training a spatial-detection neural network on a Luxonis OAK-D Pro W and Raspberry Pi Zero 2 W via the DepthAI API to flag obstacles and fast-moving hazards in under 500 ms
- Designed an MQTT (Mosquitto) pub/sub architecture relaying detections between Python services to haptic-motor and text-to-speech alerts

Private Information Retrieval (PIR)

[GitHub](#)

- Built a Path ORAM access layer in C between client applications and Apple's FoundationDB (7.1.35), obfuscating access patterns to defend against a malicious storage server (4-person team)
- Implemented a client library mirroring FoundationDB's C API plus a tree-structured ORAM server, using OpenSSL AES over fixed 1024-byte blocks so real and dummy traffic are indistinguishable; extended it to multi-client concurrent access

Houston for Humans

[GitHub](#)

- Built an interactive civic-data web app mapping Houston's walkability, transit, and flood resilience with Next.js 16 (App Router), TypeScript, and Tailwind CSS v4, isolating WebGL map logic in SSR-safe dynamic Client Components
- Rendered toggleable GeoJSON layers (walk-score gradients, existing/proposed rail & BRT, flood-risk zones) with Mapbox GL JS via react-map-gl

EDUCATION

Bachelor of Science in Computer Engineering

May 2024

Boston University, Boston, MA

Dean's List Fall 23'

LEADERSHIP EXPERIENCE

Society of Hispanic Professional Engineers | President

July 2023 - Present

- Coordinated university, corporate partners, and executive board to plan 20 events, raise \$34K, and sponsor 28 students for National Convention, the largest funding and engagement in chapter history
- Led a 12-person team and built sponsor partnerships that grew membership by 40+ students